Fiscal Year:	FY 2016	**Task Last Updated:**	FY 07/18/2017
PI Name:	Clark, Torin K. Ph.D.		
Project Title:	Predicting Sensorimotor Adaptation to Altered Gravity by Measuring Vestibular Perceptual Thresholds		
Division Name:	Human Research		
Program/Discipline:	NSBRI		
Program/Discipline--Element/Subdiscipline:	NSBRI--Sensorimotor Adaptation Team		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	torin.clark@colorado.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	303-915-2152
Organization Name:	University of Colorado, Boulder		
PI Address 1:	Smead Aerospace Engineering Sciences		
PI Address 2:	3775 Discovery Dr, Rm. AERO N301		
PI Web Page:			
City:	Boulder	**State:**	CO
Zip Code:	80303-7813	**Congressional District:**	2
Comments:	NOTE: PI moved to University of Colorado after NSBRI Postdoctoral Fellowship concluded in late 2015 (Ed., 9/1/17)		
Project Type:	Ground	**Solicitation / Funding Source:**	2013 NSBRI-RFA-13-01 Postdoctoral Fellowships
Start Date:	09/30/2013	**End Date:**	12/31/2015
No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	1	**No. of Master' Degrees:**	1
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NSBRI
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: End date changed with NSBRI report submission (Ed., 9/1/2017)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Merfeld, Daniel (MENTOR/ Massachusetts Eye and Ear Infirmary)		
Grant/Contract No.:	NCC 9-58-PF03502		
Performance Goal No.:			
Performance Goal Text:			
	POSTDOCTORAL FELLOWSHIP During and shortly after gravity transitions astronauts suffer from sensorimotor impairment which may impact critical mission tasks, particularly vehicle control and landing. While all astronauts eventually adapt to the altered gravity environments reducing sensorimotor impairment, there appear to be substantial individual differences in the severity of impairment and how long is required for adaptation, which are poorly understood. We have proposed a novel hypothesis to explain individual differences in sensorimotor adaptability: sensory noise, which we can assay using threshold measures—most relevantly vestibular perceptual thresholds—is a limiting factor in sensorimotor adaptation. We hypothesize individuals with higher noise (higher vestibular perceptual thresholds) will require a more sustained sensory conflict before adaptation is appropriate and thus take longer to adapt to an altered gravity environment.		

Task Description:	In our primary experiment, we test this hypothesis by 1) measuring an individual's vestibular perceptual thresholds (how small of a self-motion can be reliably perceived), then 2) quantifying that individual's adaptability to an altered gravity environment (adaptation in orientation perception to a hypo-gravity centripetal acceleration environment using a centrifuge), and finally 3) testing for a correlation between thresholds and adaptation rate. Our preliminary data analysis suggests there is indeed a significant relationship between thresholds and adaptation rate and our ongoing analyses are aimed at further evaluating our hypothesis. We aim to deliver a practical, operationally-relevant pre-flight predictor of sensorimotor adaptability to NASA. In addition to being well correlated with adaptation, in order for thresholds to be an operationally-relevant predictor they must A) be highly stable within an individual over time, since the pre-flight test may be weeks to months (or even years) before a critical gravity transition during flight and B) must be highly efficient (e.g. short testing time). To address item A, we have conducted an extensive study repeatedly measuring thresholds in a small population of subjects regularly for nearly one year and have found a remarkable consistency within each individual (contrast that with the ~10x variability in thresholds between individuals). To address item B, we have developed and begun validating a confidence task and fitting method for threshold measurement. Simulations and our data in 14 subjects suggest this approach may reduce the required testing time by ~5x. Future research will continue to investigate the relationship between an individual's sensory noise level and their ability to adapt to altered gravity environments, specifically aiming to extend our findings to spaceflight applications.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	There are several Earth-based impacts of the current research project. First, while predicting an individual's capacity for sensorimotor adaptation is important for astronauts on orbit, it is also critical for sensorimotor rehabilitation patients here on Earth. Being able to predict a patient's capacity for sensorimotor adaptation may allow for individualized rehabilitation plans. In fact, I have recently submitted a proposal to another funding agency exploring these hypotheses. Second, our advances in measuring vestibular perceptual thresholds may lead to a powerful diagnostic tool in the clinic. Current practices rely almost entirely on vestibular ocular reflex (VOR) responses to diagnose patients complaining of dizziness or vertigo. However, since symptoms (dizziness or vertigo) are illusory perceptions these reflex response tests may leave patients undiagnosed. Using vestibular perceptual thresholds as part of the test battery may provide improved diagnostics in the clinic here on Earth.
Task Progress:	In the last 3 months (10/1 - 12/31/2015) of this project, we have made substantial progress. First, we have tested 14 subjects' vestibular perceptual thresholds. Each subject was tested in 11 conditions: roll tilt at 0.2 Hz and 1 Hz, inter-aural translation at 0.3 Hz and 1 Hz, and roll rotation at 0.05, 0.1, 0.2, 0.5, 1, and 2 Hz. All of these threshold tests were done using our confidence task, which allows for binary or CSD fitting to estimate thresholds. These comprehensive set of tests allow us to validate and assess the efficiency of the confidence task and CSD fitting method. Finally, we are currently evaluating this extensive data set to further understand individual differences in vestibular perceptual thresholds. On the technical side, we developed our tilt perception adaptation paradigm, which created hypo-gravity levels of centripetal acceleration using a centrifuge. Recently, we implemented a subjective visual vertical (SVV) task in which subjects aligned an illuminated line, presented on a computer monitor, with their perceived vertical. This task allows us to quantify subjects' orientation perception during whole-body roll tilt in various altered gravity environments. We finalized our adaptation paradigm and tested 10 subjects. An additional 6 subjects (3 pilot subjects) were tested in various other configurations (e.g., hyper-gravity centripetal acceleration). We have completed our experiments and data collection and are currently in the process of analyses. Preliminary results suggest a negative correlation, as hypothesized, between an individual's adaptation in tilt perception and his/her vestibular perceptual threshold (roll tilt threshold at 0.2 Hz). In our collaboration with Drs. Mulavara and Bloomberg, we have implemented our threshold tests on a motion device in the Neurosciences Laboratory at Johnson Space Center (JSC) in Houston. We have collected threshold data on 15 subjects who also completed other predictor tests and several (non-vestibular) ground-based sensorimotor adaptation tasks. We have explored and analyzed the relationships between each of our different vestibular measures (thresholds, SOT dynamic posture tests, and oVEMPS tests) as a potential "basis" for predicting individual differences in spaceflight sensorimotor adaptation. With the success of this project, we have produced several publications. In the last 3 months of the project, we have published 1 journal article and have 2 more submitted and currently under review. In addition, we have 2 more journal articles in preparation for submission (one on the consistency of vestibular thresholds and one on lapse detection for threshold measurement) [Ed. note: report was submitted to NSBRI in Feb 2016]. Finally, there are several conference papers, abstracts, posters, and presentations which are attributable to this project.
Bibliography Type:	Description: (Last Updated: 11/28/2025)
Abstracts for Journals and Proceedings	Clark TK, Merfeld DM. "Vestibular Perceptual Noise and Adaptation to an Altered Gravity Environment." 2016 NASA Human Research Program Investigators' Workshop, Galveston, TX, February 8-11, 2016. 2016 NASA Human Research Program Investigators' Workshop, Galveston, TX, February 8-11, 2016. , Feb-2016
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Abstracts for Journals and Proceedings	Galvan-Garza RC, Clark TK, Merfeld DM, Bloomberg JJ, Oman CM, Mulavara AP. "Exhibition of Stochastic Resonance in Vestibular Perception." 2016 NASA Human Research Program Investigators' Workshop, Galveston, TX, February 8-11, 2016. 2016 NASA Human Research Program Investigators' Workshop, Galveston, TX, February 8-11, 2016. , Feb-2016

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Articles in Peer-reviewed Journals	Rosenberg MJ, Galvan-Garza RC, Clark TK, Sherwood DP, Young LR, Karmali F. "Human manual control precision depends on vestibular sensory precision and gravitational magnitude." J Neurophysiol. 2018 Dec 1;120(6):3187-97. https://doi.org/10.1152/jn.00565.2018 ; PubMed PMID: 30379610 , Dec-2018
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